Sahaj Bhadja

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Summary

Systems-focused Software Engineer with 2+ years building **concurrent, fault-tolerant services**, evaluation pipelines, APIs, and cross-platform automation using **C++, Python, Linux, multithreading**, testing, and performance optimization.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Reworked **fault-tolerant asynchronous services** with explicit job states, isolated failures, and production monitoring, tripling throughput for 1K+ concurrent jobs while sustaining 99%+ reliability across 5K+ monthly runs.
- Optimized multi-tenant data paths and indexed queries across PostgreSQL and SQL Server, reducing dashboard latency from 1.2 seconds to 350 milliseconds while supporting 500+ daily platform users.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built **multithreaded processing workflows** for PDF, DOCX, ZIP, OCR, and vision-model inputs, processing 1K+ files per batch while reliably reducing instructor feedback turnaround by 60% in production.
- Developed React and Flask interfaces for uploads, model selection, grading history, and feedback, serving 150+ instructors and 10K+ daily requests through **tested production APIs** reliably at scale.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a Decision Transformer for autonomous-driving offline reinforcement learning, improving trajectory planning 18% while processing 100K+ simulated driving logs through reproducible **Python evaluation pipelines** for controlled experiments.
- Implemented a spatiotemporal encoder integrating HD-map geometry, lane context, and ego trajectories with multi-head attention, consistently improving closed-loop stability 15% across **measured simulation baselines** for planning research.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Designed cross-platform integration modules using **object-oriented architecture**, reducing API response times 30%, technical debt 20%, and duplicated logic across a highly reliable, scalable, maintainable microservice-based data-sharing platform.
- Automated 20+ **unit and integration tests** with PyTest, Mockito, Jenkins, and virtual-machine builds across Windows, Linux, and macOS, collaborating effectively through Agile reviews and Git workflows worldwide.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Developed modular services with REST APIs, WebSockets, authentication, permissions, and notifications, synchronizing 2M+ distributed records while reducing staged-release conflicts 33% across **cross-platform production workflows** safely at scale.
- Implemented automated validation, Docker packaging, Jenkins CI/CD, and centralized monitoring, cutting deployment time 50% while consistently sustaining **95%+ reliability** across repeatable Windows and Linux production release environments.

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, SQL, Bash, Shell, JSON, XML, YAML
Operating Systems: Linux, Unix, Windows, POSIX APIs, Processes, Threads, Concurrency, Multithreading, Mutexes, Locks, Semaphores, IPC, Memory Management, CPU Scheduling
Systems & APIs: Event Loops, epoll, System Services, Daemons, TCP/IP, WebSockets, REST APIs, Asynchronous Programming, State Machines, Distributed Systems
Testing & Performance: Unit Testing, Integration Testing, PyTest, Mockito, Debugging, Performance Profiling, Benchmarking, Fault Isolation, Load Testing, A/B Testing
Tools & Practices: Git, GitHub Actions, Jenkins, CI/CD, Docker, Linux CLI, Agile, Code Reviews, Object-Oriented Design, System Design, Technical Documentation, AI/ML

Projects

ExperimentKit Runtime | *C++, Linux, POSIX, IPC, SQLite, CMake, GoogleTest*

- Designed an **on-device experimentation service** with deterministic assignment, local configuration caching, Unix-domain IPC, and developer-facing APIs, isolating failures while validating enrollment, persistence, and restart behavior through integration tests.

ConcurrencyProbe | *C++, Linux, epoll, Threads, Mutexes, Semaphores, Benchmarking*

- Built a **high-performance event-driven library** combining epoll, bounded worker queues, synchronization primitives, and structured telemetry, with repeatable benchmarks for latency, throughput, memory behavior, and contention under concurrent workloads.

CPU-Scheduling Simulator | *C++, React.js, JavaScript, Operating Systems, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue scheduling with **C++ algorithm logic**, interactive controls, edge-case testing, and dynamic turnaround-time visualization across 1K+ simulated process combinations.